

EEG Sleep Classification and Reference-Based Sleep-Health Profiling

Existing evidential Transformer with targeted N1/N2 spectral refinement

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Problem and project scope

Classify each 30-second EEG epoch as Wake, N1, N2, N3, or REM.

Baseline confusion: 506 true N2 epochs were predicted N1.

Extend the retained classifier with spectral refinement and recording-level descriptive profiles.

HMC comparisons are research reference comparisons, not diagnostic conclusions.

Implemented project objectives

- Retain the five-stage evidential Transformer and its uncertainty output.
- Refine Transformer-predicted N1/N2 using nine EEG spectral features.
- Calculate architecture metrics and aggregate spectral biomarkers.
- Compare with the 16-recording training reference and display four-domain deviation profiles.

HMC Sleep Staging Database v1.1

151 whole-night PSG recordings; 85 male and 66 female.

Mean age: 53.9 ± 15.4 years.

A heterogeneous population referred for PSG examinations.

The project uses 24 recordings and four EEG channels sampled originally at 256 Hz.

Recording-wise experimental split

Partition	Recordings	Stored epochs
Training	16	15,216
Validation	4	3,619
Testing	4	3,109
Total	24	21,944

Test IDs: SN001, SN004, SN009, SN022. SN identifiers name recordings; partitions are disjoint.

Evaluation scope and SN001 segment

Current notebook: 3,109 test epochs, including SN001's stored 190 epochs (95 minutes).

Earlier EDF alignment audit: SN001 reconstructed to 854 epochs; test total 3,773.

The 55.45% → 59.34% comparison belongs to the current stored-input experiment.

SN001's segment duration and onset do not establish whole-night sleep-health measurements.

EEG preprocessing and model input

Channels: EEG F4-M1, EEG C4-M1, EEG O2-M1, EEG C3-M2.

Actual preprocessing order: 50 Hz notch at the original rate, 0.3–35 Hz bandpass, resample to 100 Hz.

Complete 30-second epochs; per-epoch, per-channel normalization.

Each input epoch: 4 channels $\times$ 3,000 samples. Apply the notch before 100 Hz resampling.

Retained Fast SCFormer-U architecture

- | $4 \times 3,000$ EEG $\rightarrow$ Conv1D 64 (kernel 7, stride 4) $\rightarrow$ Conv1D 128 (kernel 5, stride 4)
- | Self-attention: 4 heads, key dimension 16; residual normalization and feed-forward layers
- | Bidirectional LSTM: 64 units per direction; pooling and dense evidence head
- | Five softplus evidence outputs; 226,309 total parameters. Attention and recurrence operate within the epoch.

Role of the retained components

Component	Role
CNN encoder	Local EEG patterns and temporal downsampling
Self-attention	Relations between positions within each epoch
BiLSTM	Downsampled within-epoch sequence
Evidence head	Five-class Dirichlet evidence
Spectral refinement	Independent classifier for N1/N2 candidates

Evidential probabilities and uncertainty

$$\alpha_k = \text{evidence}_k + 1$$

$$P_k = \alpha_k / \text{sum}(\alpha)$$

$$\text{Uncertainty} = 5 / \text{sum}(\alpha)$$

Refinement changes the selected N1/N2 label. Transformer probabilities and uncertainty still refer to the original five-stage output.

Retained training configuration

Setting	Existing Transformer
Optimizer / learning rate	Adam / 0.0001
Batch size / maximum epochs	128 / 10
Validation-selected checkpoint	Epoch 9
Class weighting	Training labels only
Architecture replacement	None

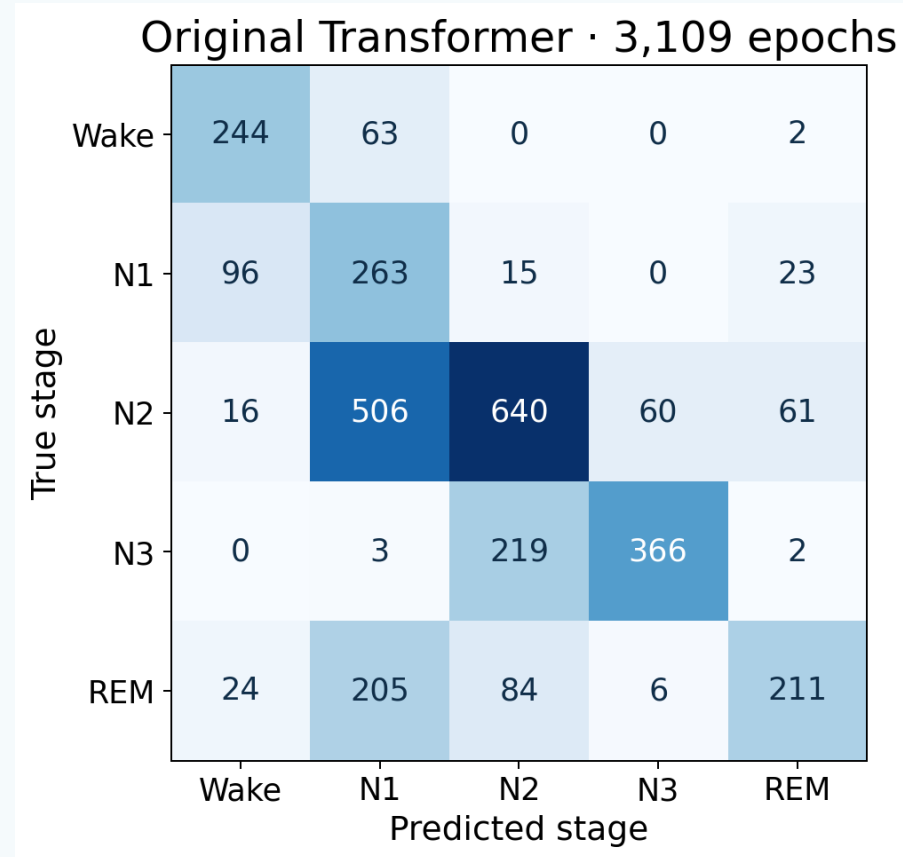
The refinement audit loads the saved checkpoint; it does not retrain the Transformer.

Original Transformer performance

Metric	Stored-input test set
Accuracy	55.45%
Macro F1	57.42%
N1 F1	36.60%
N2 F1	57.12%
Evaluated epochs	3,109

Same labels and epochs are used for the baseline/refinement comparison.

Baseline confusion matrix



Largest N2 error: 506 N2 epochs predicted N1. Rows are true labels; columns are predictions.

Nine EEG features for N1/N2 refinement

Feature group	Features
Relative band power	Delta, Theta, Alpha, Sigma, Beta
Ratios to Delta	Sigma/Delta, Theta/Delta, Alpha/Delta, Beta/Delta
Frequency bands (Hz)	0.5–4; 4–8; 8–12; 12–16; 16–30
Candidate gate	Transformer predicts N1 or N2

These are project feature boundaries, not AASM staging definitions. Sigma power does not measure spindle density.

Welch spectral extraction

Welch PSD per channel: 100 Hz, 400-sample windows (4 seconds), 200-sample overlap.

Integrate each band with a lower-inclusive, upper-exclusive frequency mask.

Average band powers across the four channels before computing relative powers and ratios.

Relative power denominator: measured total power in 0.5–30 Hz. Aggregate epoch features by arithmetic mean per recording.

Relative powers and Delta ratios

Relative band power = mean channel band power / mean channel total power

Band/Delta ratio = mean channel band power / mean channel Delta power

Delta is a project feature-engineering reference, not a mandatory denominator.

Normalized EEG powers are spectral features; do not describe them as calibrated absolute physical power.

Training and inference gates

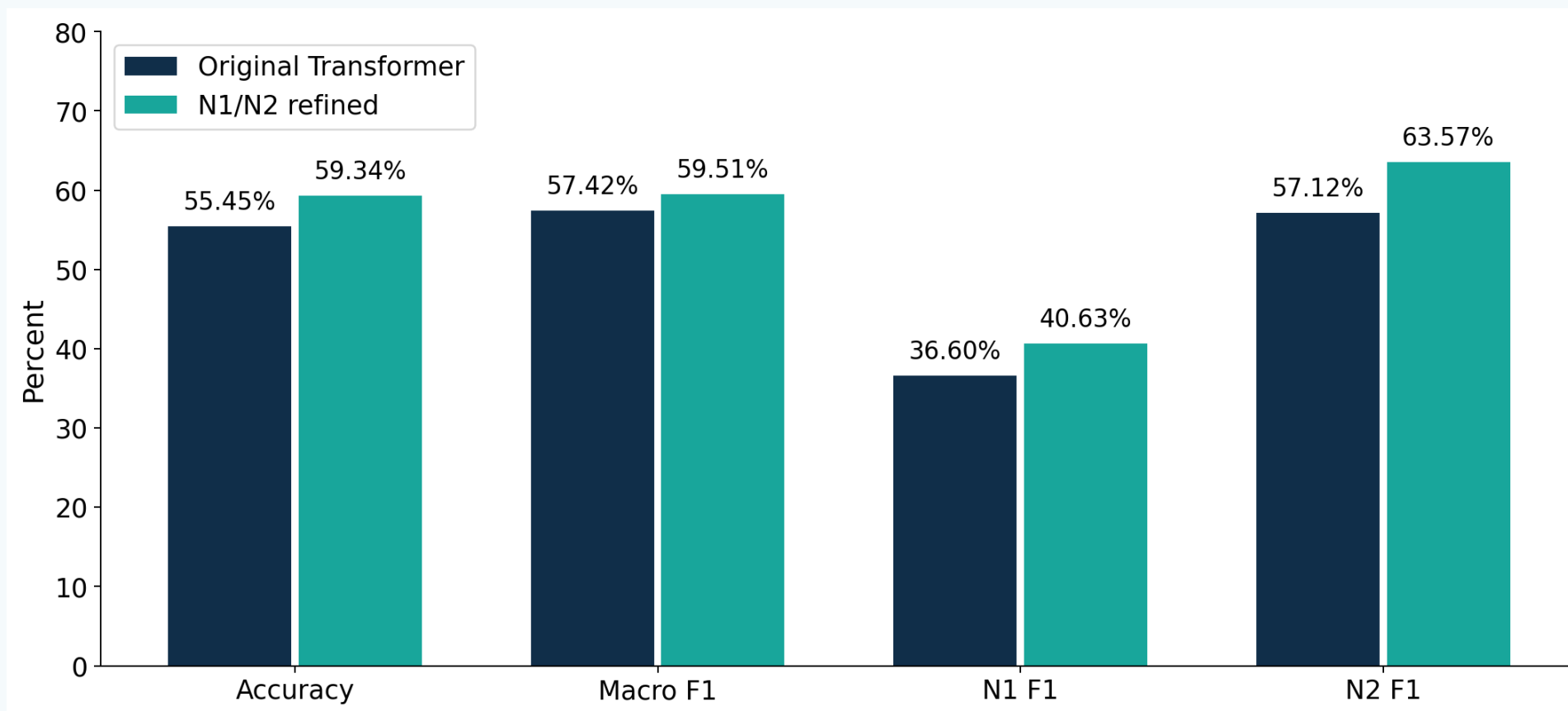
- Fit the refiner only on training epochs where the true stage AND the Transformer prediction are N1/N2.
- Training subset: 1,278 N1 and 4,588 N2 epochs; 5,866 total.
- StandardScaler → LogisticRegression(class_weight="balanced", max_iter=2000, random_state=42).
- At inference, gate by the Transformer prediction only. Ground truth is never used to route test epochs.

Standardized logistic regression coefficients

Feature	Coefficient toward N2
delta relative	0.762
theta relative	1.377
alpha relative	-0.827
sigma relative	2.135
beta relative	-1.783
sigma delta ratio	-1.436
theta delta ratio	-1.066
alpha delta ratio	0.995
beta delta ratio	0.309

Multivariate coefficients depend on correlated features and standardization; do not read them as isolated physiological rules.

Effect of targeted N1/N2 refinement



Accuracy +3.89 pp; macro F1 +2.10 pp using unrounded values. N1 F1 +4.03 pp; N2 F1 +6.45 pp.

Which predictions changed?

Outcome	Epochs
Unchanged	2,871
Corrected	154
Wrong to wrong	51
Degraded	33
Total changed	238

W, N3 and REM predictions are retained. Epochs truly in those stages can still enter refinement if originally predicted N1/N2.

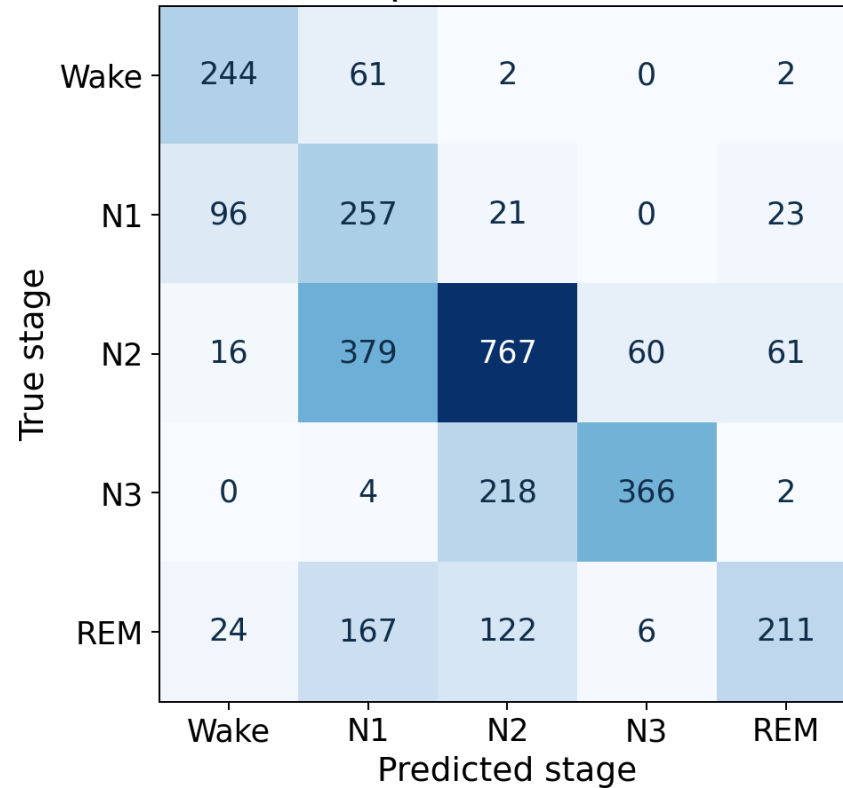
Performance by test recording

Recording	Epochs	Baseline accuracy	Refined accuracy
SN001	190	46.84%	48.42%
SN004	1,016	56.79%	64.07%
SN009	901	61.27%	64.48%
SN022	1,002	50.50%	52.00%

The SN001 evaluation covers 95 minutes. No whole-night performance or external cohort validation is implied.

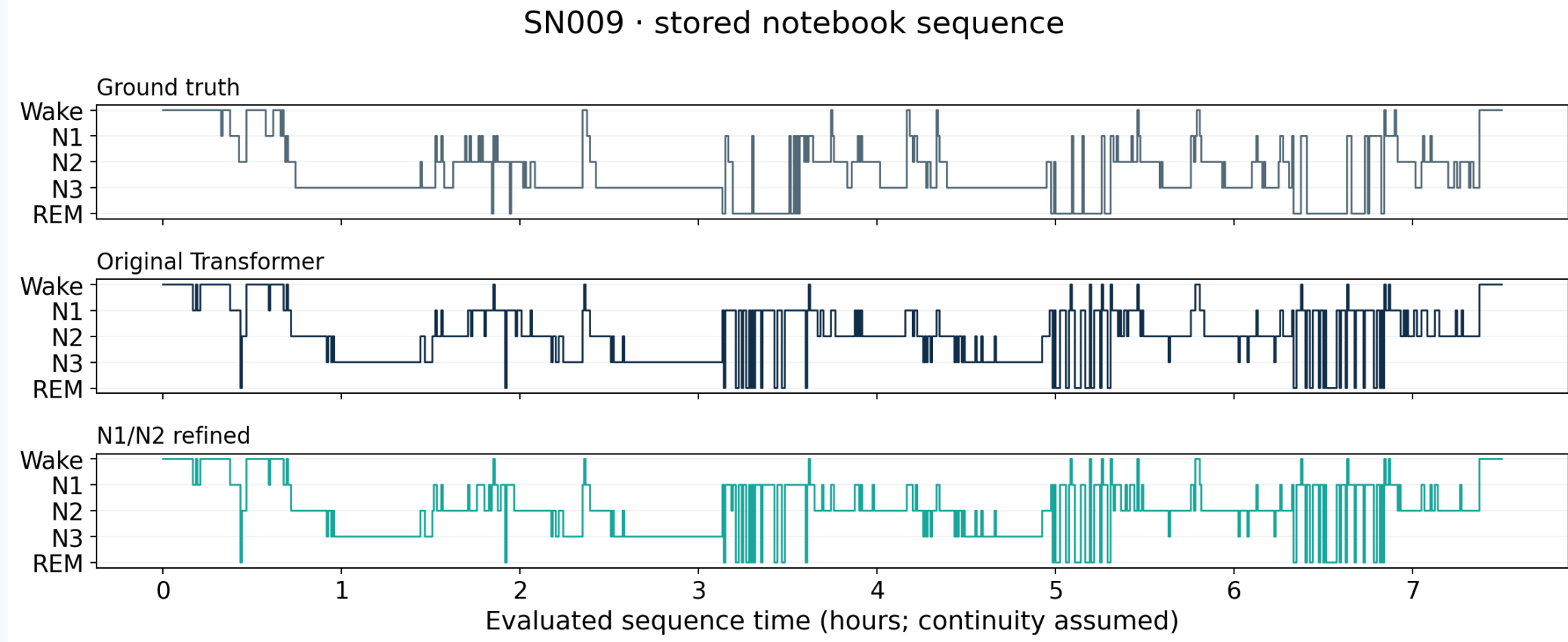
Refined confusion matrix

Transformer + N1/N2 spectral refinement · 3,109 epochs



N2→N1 errors: 506 → 379. All true N1/N2 epochs: accuracy 53.75% → 60.95% (1,680 epochs).

Original and refined hypnogram — SN009



Targeted N1/N2 changes retain the other predicted stages and the recording order.

Sleep architecture measurements

Metric	Definition
TST	$N1 + N2 + N3 + \text{REM time}$
Sleep efficiency	$100 \times \text{TST} / \text{evaluated recording duration}$
Stage percentages	$100 \times \text{stage sleep time} / \text{TST}$
WASO	Wake time after first sleep to sequence end
Fragmentation	All stage transitions / TST hours

Recording duration is an evaluated-window measure. Contiguous timing is assumed for the notebook NPZ sequences.

Sleep duration and efficiency

- | Each stored epoch contributes 0.5 minutes. Wake is excluded from TST.
- | Sleep efficiency uses the duration of the evaluated sequence, not independently verified time in bed.
- | N1/N2 refinement cannot change TST or efficiency because both labels are sleep.
- | SN001's 91.5 minutes predicted sleep in a 95-minute segment must not be treated as whole-night sleep duration.

Sleep initiation and Wake after sleep onset

SOL = index of first non-Wake epoch × 0.5 minutes.

WASO includes every Wake epoch after onset, including Wake at the end of this notebook's sequence.

REM latency = time from first sleep epoch to first REM epoch.

No sleep or no REM makes the corresponding metric unavailable. Segment onset is relative to the segment, not lights-out.

Stage percentages and fragmentation

- | N1%, N2%, N3%, REM% use TST as the denominator and sum to approximately 100%.
- | Awakenings count non-Wake → Wake transitions after first sleep.
- | Stage transitions count every adjacent pair whose labels differ.
- | Fragmentation = stage transitions / TST hours. This project-defined index is not EEG micro-arousal scoring.

Recording-level EEG spectral biomarkers

Recording	Delta	Theta	Alpha	Sigma	Beta
SN001	0.6749	0.1201	0.0911	0.0420	0.0451
SN004	0.6632	0.1348	0.0703	0.0430	0.0611
SN009	0.6323	0.1284	0.0863	0.0562	0.0682
SN022	0.6826	0.1565	0.0384	0.0254	0.0781

Means of per-epoch relative features across all evaluated epochs, including Wake. Four band/Delta ratios are also retained.

Master feature matrix and training reference

Test matrix: 4 rows × 23 columns = subject ID + 22 numeric fields.

Training reference matrix: 16 rows × 23 columns.

Numeric fields: 13 recording/architecture fields + nine EEG spectral features.

Deviation analysis excludes recording_minutes: 21 compared fields. Ten architecture metrics feed the four domain statuses.

Training reference — architecture

Metric	Mean	SD	P10	P90
TST minutes	377.81	98.53	256.25	512.50
sleep efficiency percent	78.60	14.17	59.16	93.11
sleep onset latency min	9.63	7.21	3.50	14.25
WASO minutes	88.06	61.09	27.75	179.25
N1 percent	20.33	10.59	7.63	36.58
N2 percent	40.48	14.03	24.70	57.26
N3 percent	26.22	11.39	15.55	40.99
REM percent	12.96	5.71	5.80	19.42

Sample SD uses ddof=1; empirical quantiles use linear interpolation. Reference architecture uses fitted training-recording predictions.

Training reference — EEG features

EEG feature	Mean	SD	P10	P90
delta relative	0.6674	0.0674	0.5775	0.7401
theta relative	0.1269	0.0219	0.1046	0.1500
alpha relative	0.0888	0.0448	0.0491	0.1557
sigma relative	0.0349	0.0085	0.0236	0.0465
beta relative	0.0563	0.0284	0.0354	0.0926
sigma delta ratio	0.0656	0.0221	0.0364	0.0930
theta delta ratio	0.2234	0.0677	0.1706	0.3100
alpha delta ratio	0.1985	0.1704	0.0754	0.3880
beta delta ratio	0.1148	0.0715	0.0631	0.1878

Dataset-derived descriptive boundaries from 16 training recordings. No healthy normative cohort was established.

Z-scores and P10/P90 regions

| $z = (\text{recording value} - \text{reference mean}) / \text{reference SD}$

| LOW: below P10; NORMAL: P10 through P90; HIGH: above P90.

| Z-score measures magnitude and direction; percentile regions provide a categorical reference comparison.

| NORMAL means within the dataset interval. LOW/HIGH do not mean medically bad/good. Equality at either boundary stays within reference.

Four sleep-health domains

Domain	Metrics used
Duration	TST
Continuity	Sleep efficiency, WASO, awakenings, fragmentation
Initiation	SOL
Architecture	N1%, N2%, N3%, REM%

Any LOW or HIGH metric makes its domain DEVIATED. EEG deviations are shown separately and do not determine domain counts.

Overall descriptive profile levels

Deviated domains	Profile
0	Within Reference
1	Mild Deviation
2	Moderate Deviation
3 or 4	High Deviation

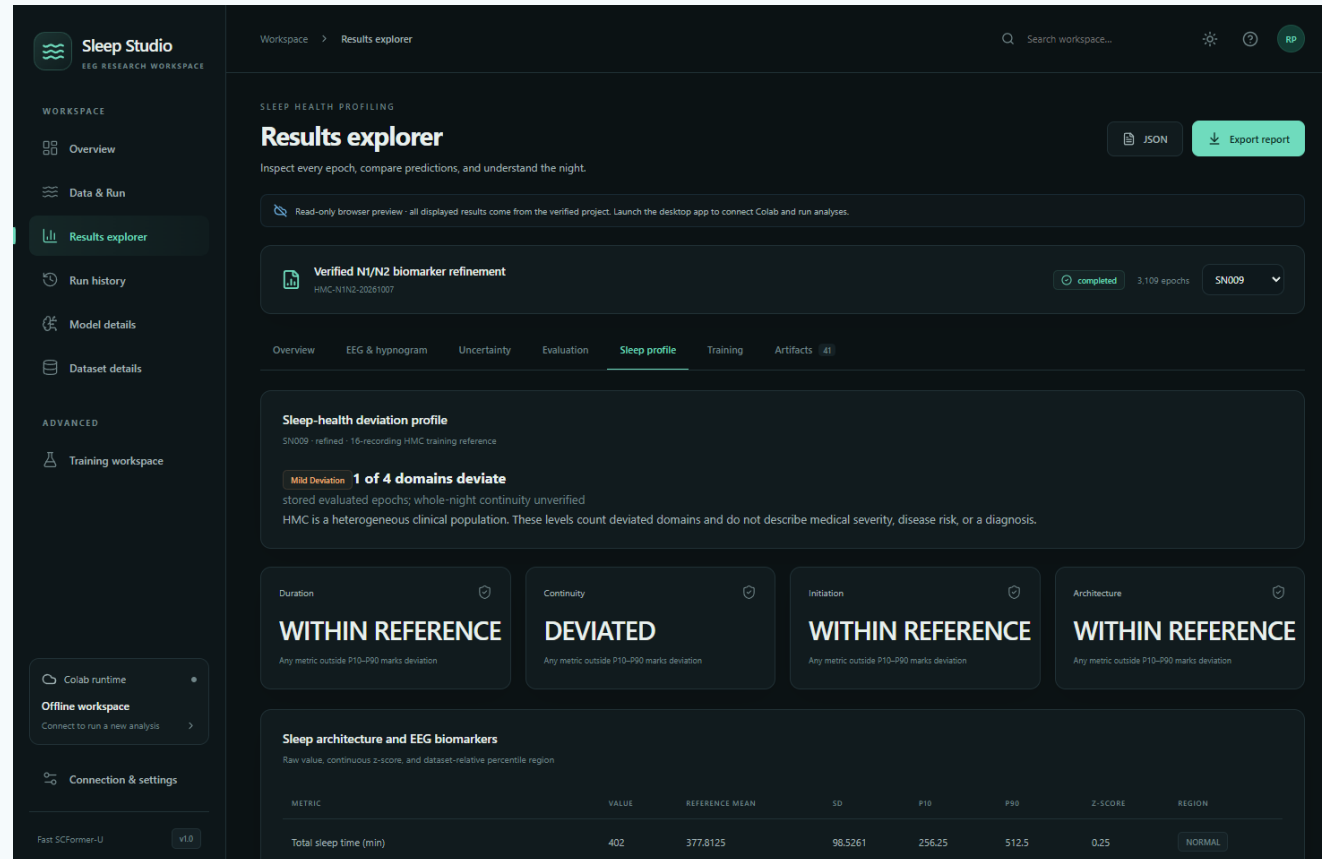
Reference boundaries describe the HMC training cohort; profile levels are descriptive domain counts.

Test-recording domain profiles

Recording	Duration	Continuity	Initiation	Architecture	Overall
SN001	Deviated	Deviated	Deviated	Within ref.	High Deviation
SN004	Within ref.	Deviated	Deviated	Deviated	High Deviation
SN009	Within ref.	Deviated	Within ref.	Within ref.	Mild Deviation
SN022	Within ref.	Deviated	Within ref.	Deviated	Moderate Deviation

SN001: segment profile. HMC reference deviations are neither disease-risk predictions nor validated severity levels.

Interactive Sleep Studio dashboard



Select SN001/SN004/SN009/SN022. Inspect domains, sleep architecture, nine EEG features, z-scores, P10/P90, and source artifacts.

Complete implemented pipeline

Raw EEG → existing preprocessing → evidential Transformer → targeted N1/N2 spectral refinement

Refined hypnogram → sleep architecture; EEG → spectral biomarkers

Recording feature matrix → 16-recording training reference → z-scores + P10/P90 regions

Four domain statuses → descriptive overall deviation profile → interactive dashboard and report exports

Limitations and unimplemented extensions

- | No explicit spindle detector, dedicated formal SWA metric, or spectral edge frequency.
- | No clinically validated normative ranges, disease prediction, or diagnostic severity classification.
- | No documented repeated-night participant mapping for true longitudinal personalization.
- | Small clinical reference (16 recordings), assumed NPZ continuity, in-sample reference architecture, and a partial SN001 input.

Model uncertainty and profile deviation

Quantity	Question answered	Source
Transformer uncertainty	How much evidence supports this epoch classification?	5 / sum(alpha)
Sleep-health deviation	How does this recording compare with the training reference?	z-score and P10/P90
Overall profile	How many architecture domains deviate?	0–4 domain count

The new label refinement does not recalibrate Transformer probabilities or provide uncertainty for the logistic classifier.

Project outcome

- The retained Transformer's test accuracy improves from 55.45% to 59.34% with targeted spectral refinement.
- Macro F1 improves from 57.42% to 59.51%; N1 and N2 remain challenging.
- The dashboard combines architecture, EEG spectral features, reference deviations, and four-domain profiles.
- Reference boundaries describe the HMC training cohort; profile levels are descriptive domain counts.

Sources and reproducibility

Implementation: Biomarker_N1_N2_Refinement.ipynb, new pipeline cells 1–46.

Independent Colab reproduction: HMC-N1N2-20261007; saved checkpoint and stored NPZ inputs.

HMC v1.1: Alvarez-Estevez & Rijsman (2022), DOI 10.13026/t79q-fr32.

Full feature tables, refiner scaler/coefficients, per-epoch outputs, hashes, and checks are in refinement_artifacts/verified.